

Adithya Sudhindra

Los Angeles, CA | (213) 301-4085 | asudhind@usc.edu | [linkedin.com/in/adithya-sudhindra](https://www.linkedin.com/in/adithya-sudhindra) | github.com/Adithya2441

Education

University of Southern California

Jan 2025 – Dec 2026

Master of Science in Computer Science, GPA: 3.74/4.0

Los Angeles, CA

- Coursework: Analysis of Algorithms, Machine Learning for Data Science, Applied NLP, Foundations of AI, Operating Systems, Distributed Systems, Information Retrieval and Web Search Engines

JSS Academy of Technical Education

Aug 2019 – Jun 2023

Bachelor of Engineering in Computer Science, CGPA: 9.24/10.0

Bengaluru, India

- Relevant Coursework: Data Structures, Digital Image Processing, Data Mining and Data Warehousing, AI and ML

Experience

Northern Lights (Presage)

Jan 2026 – Present

Machine Learning Engineer Intern

Los Angeles, CA

- Building machine learning models using Google's Unsplash dataset along with its metadata (**125GB+**), leveraging **AWS S3** for data storage and **AWS SageMaker** for training and experimentation
- Working on developing an **agentic AI model** using **Model Context Protocol (MCP)** and **AWS services** for a YouTube Shorts dataset to enable intelligent and autonomous task execution

Mercedes-Benz Research and Development India

Jul 2023 – Dec 2024

Consultant (Software Developer)

Bengaluru, India

- Spearheaded development and integration of Generative AI-powered features using **GPT-4o**, **Ada API** and **Django**, driving major enhancements in intelligent automation across core applications
- Architected and automated documentation generation within in-house platforms by leveraging GPT-4o LLM and Ada API, increasing documentation PDF engagement by **30%** and streamlining support workflows
- Formulated and executed optimized model delivery pipelines and codebase, resulting in an **18% increase** in processing efficiency through targeted refactoring and performance tuning
- Engineered user-centric features such as **image querying and voice input** to augment application support capabilities and improve user experience

Saigeware Technologies Pvt Ltd

Feb 2023 – Apr 2023

AI and Data Science Intern

Bengaluru, India

- Designed and implemented an end-to-end **ML pipeline**; developed a custom ensemble of Random Forest and deep learning models for image and sound data from 800+ clinical patients, elevating multiclass classification accuracy from **55% to 75%**
- Built data extraction pipelines for large-scale healthcare datasets, including advanced feature extraction from image data using **Variational Autoencoders (VAEs)** to learn meaningful latent representations
- Performed visualization, data extraction, and trend analysis on a US healthcare dataset of **1.2 billion** patient records; automated data workflows to reduce manual intervention by 30%

Projects

AI-Based Image Generation | *Stable Diffusion, CLIP*

- Reproduced Stable Diffusion using **80,000 images**, achieving coherence and accuracy competitive with original models (baseline CLIP score: 34.9–36.2; artist imitation **accuracy: 81%**). Fine-tuned pipeline to improve text-to-image quality by 25%

Machine Learning Projects | *Scikit-learn, Kaggle*

- Performed sentiment analysis on Amazon product reviews employing advanced machine learning algorithms (Random Forest, XGBoost, Multinomial Naive Bayes, LSTM, SVM), achieving **77–93%** accuracy and ROC-AUC up to 0.92

Waste Image Classification using Transfer Learning | *OpenCV, TensorFlow, Scikit-learn*

- Built a multi-class waste image classifier utilizing **transfer learning** (ResNet50, ResNet101, EfficientNetB0, VGG16) with data augmentation, delivering up to **84.9%** validation accuracy and 0.980 AUC

Technical Skills

Programming Skills: Python, Java, C, C++, SQL, HTML, CSS, JavaScript, R, Kotlin

Frameworks/Tools: AWS, Django, StreamLit, Jira, MySQL, PostgreSQL, MongoDB, Firebase, Docker, CI/CD, Git, Linux, CUDA

ML Libraries: NumPy, Pandas, Matplotlib, Scikit-learn, OpenCV, TensorFlow, Seaborn, Statsmodels, Pytorch, Langchain

Areas: LLMs, Machine Learning, Deep Learning, Computer Vision, Reinforcement Learning, NLP, RAG, ML Ops, MCP

Accomplishments

- Placed 5th in company-wide AWS DeepRacer autonomous AI racing competition, applying reinforcement learning techniques to optimize racing models
- Developed and pitched a novel LLM-based Streamlit chatbot solution for enhancing Mercedes-Benz car manual usability; engineered system to allow users to upload images of car components and receive contextual, manual-driven responses via GPT-4o and Ada API
- Selected among a cohort of **300** students in USC's **Machine Learning for Data Science** course for a competitive internship opportunity at **Northern Lights (Presage)**